Aranda Preliminary Network Assessment

Revised executive review version based on the supplied configs, topology, and April 9 meeting clarification

Document	Internal technical assessment
Prepared for	CEO and executive leadership review
Prepared on	April 9, 2026
Status	Preliminary - final sign-off depends on Meraki MX/MS review
Revision note	Updated to reflect the April 9 meeting clarification that routing is believed to reside on the Meraki MX firewall.

Confidentiality note: This assessment is intentionally labeled preliminary. It is suitable for executive review and client discussion, but it should not be represented as a final end-to-end sign-off until the Meraki MX and Meraki MS configuration set is reviewed.

1. Executive summary

Bottom line: Current state: the environment appears to be functionally segmented and largely Layer 2 at the switch edge, but it carries meaningful security and supportability gaps. The largest remaining blind spot is the Meraki MX/MS layer, which likely owns live routing and core switching policy.

Category	Executive view
Overall posture	Moderate to high risk until the Meraki MX/MS layer is reviewed.
What is working	The network appears to be segmented by business function, and the non-Meraki office switches are generally operating as Layer 2.
What is missing	Core Meraki switching and the MX firewall configuration are not yet part of the evidence set, so end-to-end policy validation is incomplete.
What requires immediate attention	Sensitive keys and certificates embedded in the exported backups, plus local account sprawl and management-plane exposure.

The supplied non-Meraki switch exports do provide enough evidence to identify immediate risk items. The most urgent concerns are the presence of private keys and certificate material inside the backups, inconsistent privileged local account hygiene, management-plane exposure across multiple VLANs, and incomplete trunk standardization.

2. What changed in this revised version

- The revised assessment is aligned to the April 9 meeting clarification that the switch estate is operating primarily as Layer 2 and that inter-VLAN routing is believed to reside on the Meraki MX firewall.
- The revised scope explicitly treats the Meraki MX and Meraki MS estate as the most important remaining data gap, because those systems determine routing, core switching policy, and much of the segmentation enforcement.
- The revised deliverable separates the client-facing proposal from the technical assessment so leadership can review a polished statement of work and an internal risk register independently.

3. Scope clarification from the April 9 meeting

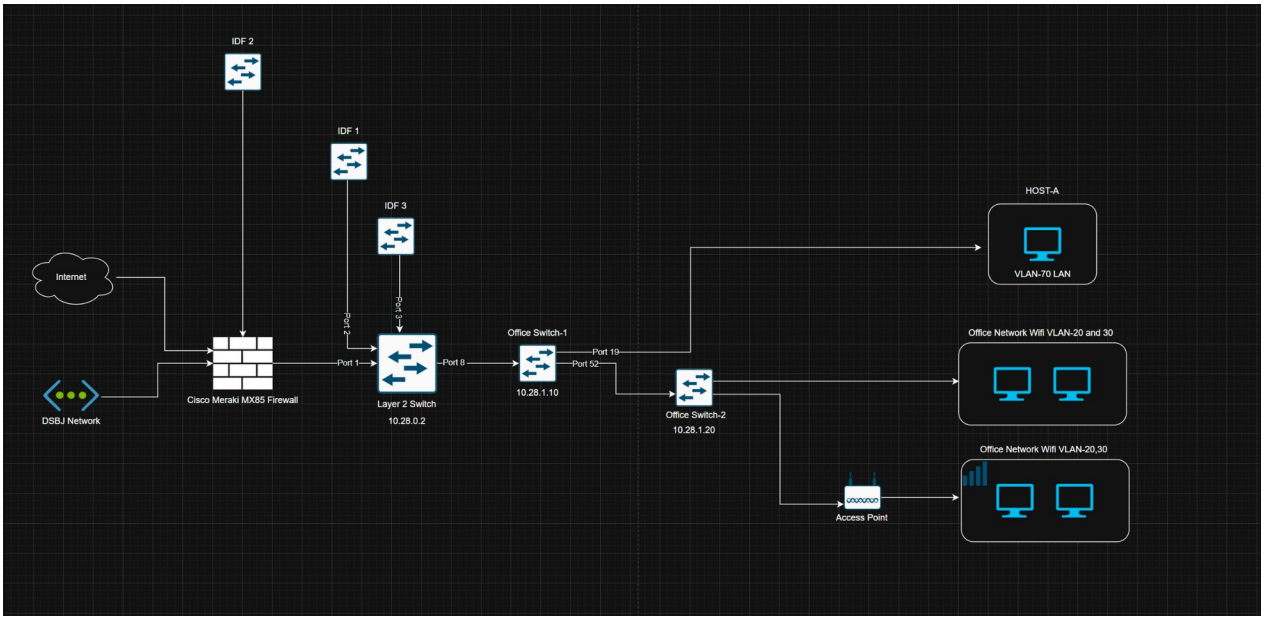
Why this matters: The internal meeting clarified three important points: (1) the Meraki MS switch estate was not part of the exported backup bundle, (2) the switch estate is believed to be operating primarily as Layer 2, and (3) the Meraki MX firewall is believed to own routing. Those clarifications materially change where the final assessment effort must focus.

The practical effect is straightforward: the current evidence set is good enough to assess the non-Meraki edge and access-layer posture, but not sufficient to validate the live control plane, routing, or end-to-end enforcement points. That final validation requires Meraki Dashboard/API review.

4. Environment summary and source material

Component	Status in this review	Observation	Why it matters
Meraki MX85 firewall	Logical perimeter and routing point	Present in client-supplied topology; live config not yet exported.	Critical missing data source because routing and policy enforcement are believed to reside here.
Meraki MS switch layer	Core / aggregation switching	Explicitly excluded from client backup set.	Likely includes the 10.28.0.2 aggregation function shown in the topology and must be reviewed before final sign-off.
arandaswitch01	10.28.1.10 / 10.10.0.10 / 172.28.5.5 / 172.28.12.5	Text config reviewed	Likely office access/distribution switch (matches Office Switch-1 management IP from topology).
arandaswitch02	10.28.1.20 / 10.10.0.20	Text config reviewed	Likely office access/distribution switch (matches Office Switch-2 management IP from topology).
arandaswitch03	10.28.1.30 / 10.10.0.30	Text config reviewed	Small access switch or IDF edge role.
arandaswitch04	10.10.0.40	Text config reviewed	Minimal config; validate whether active, spare, or incomplete export.
Netgear smart switches	10.10.0.202-215 subset observed	Mixed text and binary exports reviewed	Represents additional edge switching; several exports are not human-readable without native tools.

The client-supplied topology is useful context, but it must still be treated as unvalidated until the Meraki Dashboard is reviewed. The image below is best understood as the intended design, not yet the fully confirmed design.



Client-supplied logical topology - useful context, but not yet validated against the Meraki Dashboard.

5. Observed VLAN architecture from the supplied configs

A positive finding is that the client appears to have an intentional segmentation model. The office, guest, IoT, printer, and OT zones are named consistently across the main office switch exports. The open question is not whether segmentation exists in theory, but whether it is enforced correctly at the Meraki control points.

VLAN	Observed purpose	Assessment note
10	Office Management	Defined on arandaswitch01 and arandaswitch02.
20	Office Wired / Office Computer	Primary office user VLAN across the non-Meraki access layer; also seen on Netgear15.
30	Office Wireless	Defined in core office switch exports; management IP also present on arandaswitch01.
40	Office Wi-Fi Guest	Defined in office switch exports; enforcement still needs Meraki validation.
50	Office Services	Defined in office switch exports.
60	Office IoT	Defined in office switch exports.
70	Office Printers	Defined in office switch exports; management IP also present on arandaswitch01.
100	OT Management	Defined consistently and assigned management IPs on arandaswitch01, 02, and 03.
110	OT Computers	Defined in office switch exports.
120	OT Printers	Defined in office switch exports.

6. What can be stated with confidence today

Device / source	Observed technical state	Why it matters
arandaswitch01	no ip routing; VLANs 10/20/30/40/50/60/70/100/110/120; switch IPs on VLAN1, 30, 70, and 100; STP priority 4096; trunk uplinks present.	Supports the Layer 2 operating model but also shows broad management-plane exposure and incomplete trunk standardization.
arandaswitch02	no ip routing; VLANs 10/20/30/40/50/60/70/100/110/120; switch IPs on VLAN1 and 100; trunk uplinks present.	Confirms consistent segmentation intent across the main office switches but still requires Meraki-side validation for end-to-end enforcement.
arandaswitch03	Small access switch with VLANs 20 and 100 only; one trunk uplink.	Likely an edge or IDF role with simpler access-layer function.
arandaswitch04	Management IP plus embedded key/certificate material; very limited additional config in the supplied export.	Requires field validation because its operational role is not clear from the backup alone.

Current confidence: The main office switches appear to be operating as Layer 2, which is directionally consistent with the meeting note that the Meraki MX firewall owns routing. That is an important clarification, because it means final segmentation validation depends more on the MX/MS review than on the non-Meraki access-switch review.

7. Risk-ranked findings

ID	Severity	Finding summary
F1	Critical	Sensitive cryptographic material is embedded directly in the exported switch backups.
F2	High	The Meraki MX and Meraki MS layer is the largest remaining blind spot, so the current assessment cannot yet validate the live routing and core switching policy.
F3	High	The management plane is spread across multiple VLANs, including user-adjacent networks, rather than being tightly confined to one dedicated management zone.
F4	High	Trunking standards are incomplete and do not show explicit VLAN pruning on the supplied office switches.
F5	Medium	Layer 2 access-port guardrails are not evident in the provided non-Meraki exports.
F6	Medium	Local administrative access is not standardized and includes shared or role-inappropriate privileged accounts.
F7	Medium	Operational standards are inconsistent across the switch estate.
F8	Medium	The switching estate is heterogeneous and only partially machine-readable, which reduces supportability.
F9	Low	At least one switch export appears minimally configured and should be validated against the actual site role.

F1 - Sensitive cryptographic material is embedded directly in the exported switch backups.

Priority: Critical severity

Observed state: Several provided switch exports include SSH private keys and certificate private keys, not just ordinary interface or VLAN settings. Because these backups have already been zipped and circulated, they must be treated as sensitive secrets rather than normal configuration files.

Business risk: If the backup bundle is stored insecurely or shared too broadly, an unauthorized party may be able to impersonate device identity, undermine management trust, or retain privileged access long after passwords are changed.

Recommended action: Restrict access to the current backup bundle immediately. Rotate local administrative passwords, regenerate SSH keys, replace device certificates where practical, and store future configuration exports in an encrypted repository with named access controls.

F2 - The Meraki MX and Meraki MS layer is the largest remaining blind spot, so the current assessment cannot yet validate the live routing and core switching policy.

Priority: High severity

Observed state: The meeting note clarifies that routing is believed to be on the Meraki firewall and that the provided switch set excluded the Meraki MS estate. The client topology also shows a 10.28.0.2 Layer 2 aggregation function that is not represented in the supplied non-Meraki text exports.

Business risk: Without the MX and MS data, the assessment cannot fully confirm trunk pruning, inter-VLAN routing, guest isolation, office-to-OT boundaries, access-point uplink policy, or the true enforcement point for security controls.

Recommended action: Treat the current report as preliminary. Complete the Meraki data-collection step before presenting any final sign-off conclusions to the client, and explicitly state in the proposal that the MX/MS review is required to close the assessment.

F3 - The management plane is spread across multiple VLANs, including user-adjacent networks, rather than being tightly confined to one dedicated management zone.

Priority: High severity

Observed state: The supplied switch set shows management reachability on VLAN1/10.10.0.0 management addresses and, in several cases, additional switch IPs on OT Management VLAN 100. On arandaswitch01, management IPs are also present on Office Wireless VLAN 30 and Office Printers VLAN 70, which increases exposure beyond a single trusted management segment.

Business risk: This design increases attack surface, complicates troubleshooting, and makes it harder to prove that only privileged administrators can reach the management plane.

Recommended action: Collapse management to a single secured management VLAN where possible. Remove management IP exposure from user, printer, and wireless VLANs unless there is a documented operational dependency, and enforce management access through MX/MS policy and administrative ACLs.

F4 - Trunking standards are incomplete and do not show explicit VLAN pruning on the supplied office switches.

Priority: High severity

Observed state: Multiple uplinks are configured as trunks in the non-Meraki exports, but the provided snippets do not show explicit allowed-VLAN restrictions. Some interfaces also carry mixed access/native/trunk directives, which suggests organic growth rather than a tightly standardized switch template.

Business risk: When trunks are not pruned to required VLANs, unnecessary VLAN propagation increases blast radius, complicates segmentation validation, and makes accidental exposure between office, guest, IoT, printer, and OT domains more likely.

Recommended action: Review every uplink and AP-facing trunk once the Meraki data is collected. Standardize native VLAN behavior, prune every trunk to required VLANs only, and document the intended VLAN map per uplink.

F5 - Layer 2 access-port guardrails are not evident in the provided non-Meraki exports.

Priority: Medium severity

Observed state: The reviewed text exports do not show clear evidence of BPDU Guard, Root Guard, PortFast, 802.1X, DHCP snooping, or Dynamic ARP Inspection on the non-Meraki switches. The environment may rely on Meraki for some of these controls, but that has not yet been confirmed.

Business risk: Absent access-layer guardrails, the environment is more vulnerable to accidental loops, rogue switching, unauthorized endpoint attachment, and common Layer 2 abuse patterns.

Recommended action: As part of the Meraki/MS review, confirm which Layer 2 protections are enforced centrally and which must be enabled on non-Meraki access devices. Standardize guardrails on edge ports wherever the platform supports them.

F6 - Local administrative access is not standardized and includes shared or role-inappropriate privileged accounts.

Priority: Medium severity

Observed state: The exports show multiple privilege-15 local users with inconsistent naming, including generic, vendor-like, or role-inappropriate identities. This makes it difficult to demonstrate clean joiner/mover/leaver control and administrative accountability.

Business risk: Shared or stale privileged accounts weaken accountability, complicate offboarding, and increase the chance that old credentials remain usable after personnel or vendor changes.

Recommended action: Move to named administrative accounts, reduce the number of local break-glass credentials, rotate all current local passwords, and align Meraki access to named accounts with MFA and least privilege.

F7 - Operational standards are inconsistent across the switch estate.

Priority: Medium severity

Observed state: Time services, timezone settings, service exposure, and general configuration hygiene are not consistent across the reviewed exports. Only one of the main office switches clearly shows external time-source configuration, and the provided configs do not show a standardized logging, AAA, or monitoring pattern.

Business risk: Inconsistent operational standards reduce troubleshooting quality, weaken forensic value, and make support outcomes depend more heavily on tribal knowledge than on a stable baseline.

Recommended action: Publish a standard switch baseline that covers NTP, logging/monitoring, administrative banners, credential model, port descriptions, backup handling, and approved management-plane services.

F8 - The switching estate is heterogeneous and only partially machine-readable, which reduces supportability.

Priority: Medium severity

Observed state: The evidence set includes multiple switch families and at least two different export formats. Some Netgear edge-switch exports are human-readable, while others are proprietary binary files that are harder to audit and automate against.

Business risk: Mixed platforms and inconsistent export formats increase support overhead, complicate standardization, and reduce the speed and confidence of future audits.

Recommended action: At minimum, standardize backup method, inventory, and documentation across the estate. Longer term, consider a lifecycle plan that reduces platform sprawl at the access edge.

F9 - At least one switch export appears minimally configured and should be validated against the actual site role.

Priority: Low severity

Observed state: arandaswitch04 shows a management IP and privileged access material but little else in the supplied export.

Business risk: A minimally documented or partially configured switch may be an active device, a spare, or an incomplete backup. In any of those cases, supportability and risk decisions should not be based on assumption.

Recommended action: Validate whether the device is active, where it is physically located, and whether the export is complete. If it is unused, remove or decommission it cleanly; if active, baseline it to the standard template.

8. Positive observations

- A meaningful segmentation taxonomy already exists: office management, wired users, office wireless, guest Wi-Fi, services, IoT, printers, OT management, OT computers, and OT printers are all represented in the supplied configuration set.
- The non-Meraki office switches are broadly consistent with a firewall-centric routing model; the main office switches explicitly show no IP routing, which is directionally consistent with the meeting note that routing is handled by the Meraki MX.
- The client has already exported a meaningful portion of the switching estate, which shortens discovery time and allows configuration-based analysis to begin immediately.
- The supplied topology gives enough context to trace the intended traffic flow between the MX firewall, aggregation layer, office switches, access point, and VLAN consumers, even though it still needs validation against the Meraki Dashboard.

9. Prioritized remediation roadmap

Immediate actions (0-7 days)

- Quarantine the current backup bundle and treat it as sensitive because it contains private keys and certificate material.
- Rotate local switch credentials and plan to regenerate SSH keys and certificates on devices whose private material was exposed in the export set.

- Obtain read-only Meraki Dashboard/API access for the MX firewall and MS switch estate so the assessment can move from preliminary to final.

Short-term actions (within 30 days)

- Standardize management-plane design to one secured management zone wherever operationally possible.
- Prune trunks to only required VLANs and review native VLAN handling on uplinks, AP ports, and any inter-switch links.
- Rationalize local administrator accounts and remove generic or stale privileged identities.
- Standardize NTP, logging, monitoring, port descriptions, and configuration backup handling across all switch platforms.

Stabilization and standardization (within 60-90 days)

- Complete an end-to-end segmentation review using the Meraki MX/MS/MR data set and validate guest, office, IoT, printer, and OT enforcement at the actual policy points.
- Normalize documentation: validated topology, device inventory, uplink map, VLAN matrix, and administrative access model.
- Develop a platform-lifecycle strategy for legacy or edge switches that are difficult to audit or standardize.

10. Data still required to close the assessment

Required to finish: Because routing is believed to reside on the Meraki MX and the omitted Meraki MS layer likely includes the core switching policy, the following data is not optional if the goal is a final sign-off assessment.

- Meraki MX firewall: Addressing and VLANs, appliance ports, static routes, L3/L7 firewall rules, DHCP, WAN uplinks, site-to-site VPN, and any SD-WAN or content-filtering policies relevant to segmentation.
- Meraki MS switches: switch settings, per-port configuration, link aggregations, access policies, ACLs, DHCP server policy / ARP inspection settings, and any Layer 3 interfaces if enabled.
- Meraki MR wireless (if in use): SSID-to-VLAN mapping, local-LAN access policy, and SSID firewall rules where wireless segmentation matters to the office and guest design.

A companion extraction script is included with this deliverable set so the delivery team can pull a structured Meraki assessment bundle as soon as a read-only API key is available.

11. Source materials reviewed

Source	What it contributed
Client message	Client confirmed that switch backups were exported except for the Meraki MS switches.
Client topology image	Logical topology showing MX85 firewall, aggregation switch layer, office switches, AP, and VLAN consumers.
Internal meeting note	Clarifies that routing is believed to reside on the Meraki firewall and that the switches are primarily Layer 2.
Non-Meraki text exports	arandaswitch01.txt, arandaswitch02.txt, arandaswitch03.txt, arandaswitch04.txt.
Netgear edge-switch exports	GS108Tv3 - ArandaNetgear14.cfg, GS108Tv3 - ArandaNetgear15.cfg, plus binary smart-switch exports reviewed at a limited level.